Case question framework

From The Data Science Interview Playbook by Mathew Kariuki. Print it, or keep it beside you while you practise Chapters 10 and 11.

1. Clarify the goal (2 minutes)

- What is the product or feature for? Who uses it?
- What decision depends on the answer?
- Any constraints: time, budget, data available?

Notes:

2. Structure the problem (3 minutes)

Pick one structure and say it out loud.

Case type	Structure
Measure a feature	Goal → adoption → direct effect → primary metric → long term → guardrails
Metric dropped	Is it real? → decompose → segment → hypotheses → test → recommend
North Star	Value to customers → leads to revenue → teams can move it → weaknesses
Two metrics disagree	Combine them → explain the mechanism → check profitability
No experiment possible	Comparison group (diff-in-diff) → bunching → interrupted time series
Market sizing	Top-down chain of factors → bottom-up sanity check
Unit economics	Revenue per unit → variable costs → contribution → levers
New market	Attractive? → profitable? → operable? → risk and pilot

Notes:

3. Prioritise (1 minute)

Which part would you look at first, and why?

4. Get specific (5 to 10 minutes)

Metric or analysis	Exact definition	Data source

Guardrails (what must not get worse):

5. Close the loop (2 minutes)

- Recommendation:
- What result would change your mind:
- Next step or test:

Checks before you finish

☐ Did I restate the goal before solving?
☐ Did I name one primary metric and at least two guardrails?
☐ Did I separate correlation from causation?
☐ Did I name my most sensitive assumption?
☐ Did I end with a recommendation, not "more research is needed"?